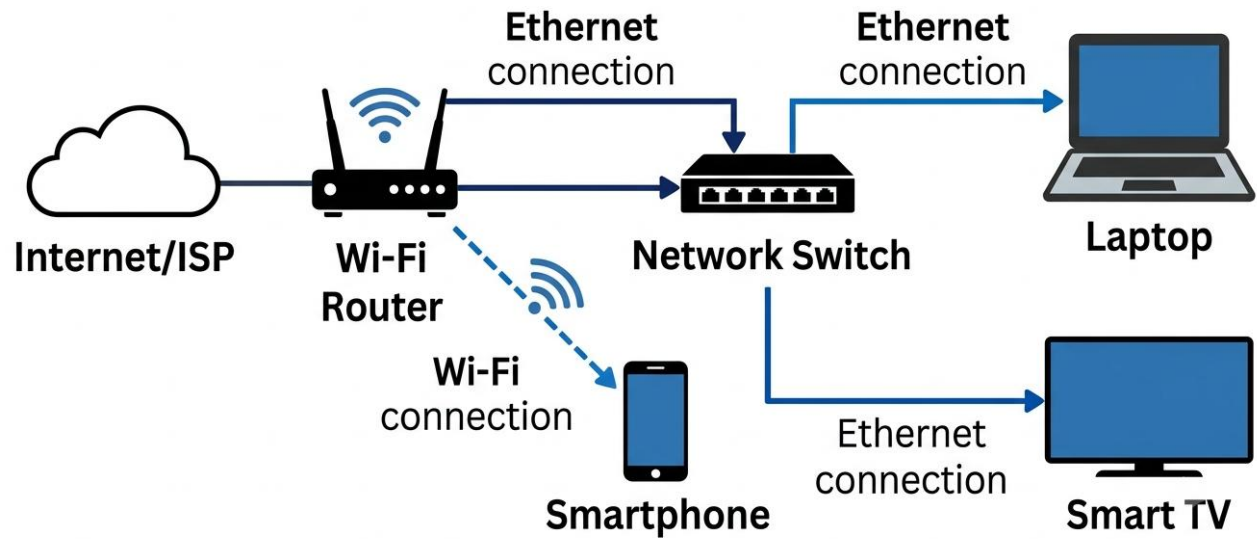


Network Devices

Task 1: Simple Home Network Diagram



Task 2: Comparison of Network Devices

Network Device	Real-world Use	Example
Router	Connects the home network to the internet.	Used for browsing, WhatsApp and Zomato.
Switch	Connects multiple wired devices in a LAN.	Used for LAN gaming and connecting PCs in a lab.
Hub	Connects multiple wired devices and sends data to all ports.	Can be used in a simple small LAN for basic file sharing.
Access Point	Provides Wi-Fi access to devices.	Used for connecting phones to Wi-Fi for YouTube streaming.

Task 3: Ordering Food on Zomato

When I order food on Zomato using my phone, the phone first connects to the Wi-Fi access point/router. The router sends my request to the ISP and then to the Zomato servers on the internet.

If a switch is used in the network, it forwards the data between wired network devices. The access point provides the wireless connection to my phone, while the router connects my home network to the internet. The Zomato server processes my order and sends the response back through the same network path.

Task 4: Connecting a Gaming PC Far from the Wi-Fi Router

1. Powerline Adapter: I can connect one powerline adapter to the router using Ethernet and plug it into a power socket. I can plug the second adapter near my gaming PC and connect it to the PC using an Ethernet cable. The internet data travels through the home's electrical wiring.
2. Wi-Fi Extender with Ethernet Port: I can place a Wi-Fi extender between the router and the gaming PC. The extender receives the Wi-Fi signal from the router and provides an Ethernet port. I can connect the gaming PC to this port using an Ethernet cable.